

Norm Regularization

$$\min: f(x) + \|Ax - b\|$$

$$\max: -f^*(-A^T z) - b^T z$$

$$\text{st: } \|z\|_X \leq 1$$

exercise - check.

Suppose f strongly convex, hence f^* smooth.

$$\hat{x} = \operatorname{argmin}_x: f(x) + z_t^T A x$$

$$z_{t+1} = \operatorname{Proj}_{\|z\|_X \leq 1} : \left(z_t + \lambda (A \hat{x} - b) \right)$$

More Norm Regularization

$$\min: f(x) + \sum_{i=1}^p \|B_i x\|_2$$

$$\max: -f^* (-B_1^T z_1 - B_2^T z_2 - \dots - B_p^T z_p)$$

$$\text{st: } \|z_i\|_2 \leq 1, \quad i=1, \dots, p.$$

$$\hat{x} = \arg\min: f(x) + \left(\sum B_i^T z_i\right)^T x$$

$$z_{i,t+1} = \text{Proj}_{\| \cdot \|_2 \leq 1} : z_{i,t} + \lambda B_i^T \hat{x}.$$

Intersection of Convex Sets - 1/3

$$\min: f(x)$$

$$\text{st: } x \in C_1 \cap C_2 \cap \dots \cap C_p$$

$$g(y_1, \dots, y_p) = I_{C_1}(y_1) + \dots + I_{C_p}(y_p)$$

$$\text{want: } g(Ax), \quad A = \begin{bmatrix} I_d \\ \vdots \\ I_d \end{bmatrix}$$

$$\text{Dual: } \max: -f^*(-z_1 - z_2 - \dots - z_p) - \underbrace{\sum I_{C_i}^*(z_i)}$$

Again assume: f^* smooth

Prox mapping of

$$\text{Use: } \text{prox}_{\lambda I_{C_i}^*}(z) = z - \lambda \text{Proj}_{C_i}(z/\lambda)$$

Intersection of Convex Sets - 2/3

Prox Grad:

$$\text{Let } \bar{z} = \sum z_i$$

$$z_{i,t+1} = z_i + \lambda \underbrace{\nabla f^*(-\bar{z}_t)} - \lambda \text{Proj}_{C_i} (z_{i,t} / \lambda + \nabla f^*(-\bar{z}_t))$$

$$\nabla f^*(-\bar{z}_t) = \underline{\hat{z} = \argmin : f(x) + \bar{z}_t^T x}$$

Intersection of Convex Sets - 3/3

Special case: Projection onto $\cap C_i$

$$\begin{aligned} \min: & \frac{1}{2} \|x - y\|_2^2 \\ \text{st: } & x \in \cap C_i \end{aligned}$$

$$0 = \hat{x} - y + \sum z_i : \hat{x} = y - \sum z_i$$

$$z_{i,t+1} = z_i + \hat{x}/p - \frac{1}{p} \text{Proj}_{C_i}(p \cdot z_{i,t} + \hat{x}) \quad // \quad \lambda = 1/p.$$

$$\text{Let: } w_i = p \cdot z_i$$

$$\hat{x} = y - \frac{1}{p} \sum w_{i,t}$$

$$w_{i,t+1} = w_{i,t} + \hat{x} - \text{Proj}_{C_i}(w_{i,t} + \hat{x}).$$